

PATIENT INPUT: Demographics, Symptoms, Vitals, Medical History

TRI-X Framework Architecture

TRIAGE

Clinical Guidelines

- €□ ACEP Dizziness/Vertigo
- €□ AHA/ASA Stroke
- €□ AAO-HNS BPPV
- €□ Red Flag Detection

TiTrATE

Diagnostic Framework

- €□ Symptom Patterns
- €□ Risk Factor Scoring
- €□ Temporal Analysis
- €□ Comorbidity Assessment

XAI

Explainability Layer

- €□ SHAP Values
- €□ LIME
- €□ NMF Phenotypes
- €□ Counterfactuals
- €□ Rule Extraction

SRGL (Screening-First Risk Governance Logic)

Gate G1: Critical Red Flags □ ' Gate G2: Risk Factors □ ' Gate G3: Uncertainty Quantification

DECISION OUTPUT

Risk Tier (R1-R5) + Urgency Level + Explanation + Care Pathway + Confidence Score

DRAS-5 States

5 Decision-Risk-Action States

ORASR Routing

Safety Routing & Care Pathways

Transparent □€□ Auditable □€□ Safety-First □€□ Clinically-Grounded